

Comments to the Authors

This manuscript addresses a timely and important question for the quantum computing community: how to identify regimes where fault-tolerant quantum computing may provide practical advantages over classical computing when evaluated not only in terms of runtime, but also in terms of space and energy consumption. The authors present a substantial amount of technical work, including logical-level resource estimation, fault-tolerant resource estimation using surface-code-based methods, and energy modeling for superconducting quantum hardware applied to the HHL/QLSA framework. The quantum algorithm analysis appears thorough, technically sophisticated, and carefully executed. The manuscript also addresses a topic that is becoming increasingly important as the community transitions from asymptotic complexity arguments toward practical assessments of quantum advantage.

Overall, I believe the paper contains the foundation of a valuable contribution. However, I also believe the manuscript currently undersells what may be its most significant scientific contribution and would benefit from substantial revision before publication.

Major Comment 1: The manuscript's strongest contribution is the framework, not the specific HHL case study

The manuscript is presently framed primarily as a resource-analysis study of the HHL quantum linear systems algorithm. However, after reading the paper, I found the more compelling contribution to be the broader methodology developed by the authors.

In particular, the paper combines:

- algorithmic resource estimation,
- fault-tolerant compilation and error correction,
- hardware-level assumptions,

- energy consumption modeling, and
- classical performance comparisons

into a single workflow intended to identify potential regions of practical quantum advantage.

This framework-oriented contribution appears more broadly impactful than the specific HHL analysis itself. The manuscript repeatedly emphasizes identifying practical quantum advantage boundaries and mapping regions of space-time-energy tradeoffs. As a result, I believe the paper would benefit significantly from elevating this framework to a more prominent position in the narrative.

I am not suggesting that the paper cease to be an HHL study. Rather, I believe the authors may be underselling what is arguably their most important contribution. The HHL analysis can serve naturally as a case study demonstrating the methodology.

A revised framing along these lines would also strengthen the paper's broader relevance to researchers interested in practical quantum advantage beyond QLSA.

Major Comment 2: The energy model needs greater robustness and sensitivity analysis

The energy analysis is one of the most interesting aspects of the paper, but it is also the area where I have the greatest concerns.

A significant portion of the conclusions rely on assumptions regarding the energy cost per physical qubit. While the manuscript appropriately acknowledges that these estimates are architecture-dependent and subject to change, many of the numerical conclusions are nevertheless presented using specific fixed values.

My concern is that future architectural improvements could materially alter these assumptions. For example, changes in cryogenic design, qubit packaging density, control architectures, or cooling-system efficiency could significantly affect the effective energy cost per physical qubit.

Because of this, I strongly encourage the authors to treat energy-per-qubit as a first-class model parameter rather than as a fixed quantity.

More importantly, I believe the paper would benefit substantially from a sensitivity analysis that explores the stability of the reported quantum advantage boundaries under perturbations of key assumptions.

Examples include:

- energy cost per physical qubit,
- physical error rate,
- logical cycle time,
- fault-tolerance overhead assumptions,
- condition number κ ,
- sparsity s ,
- precision ϵ .

At present, many of the figures represent point estimates. It is difficult to assess how robust the resulting quantum advantage regions are with respect to uncertainty in these assumptions.

A systematic sensitivity analysis would significantly strengthen the credibility and long-term relevance of the conclusions.

Major Comment 3: Practical benchmarking should be more closely tied to scientific computing practice

I do not object to the use of conjugate gradient (CG) as a classical comparison. CG is a natural algorithmic counterpart to QLSA and provides a useful theoretical baseline.

However, the manuscript frequently discusses practical quantum advantage in the context of real-world scientific computing. In that context, I believe the benchmarking discussion should be grounded more carefully in contemporary high-performance computing practice.

For example:

- Which scientific computing workloads motivate the selected parameter regimes?
- How do representative values of κ , s , and ε compare with those encountered in production HPC simulations?
- How might the conclusions change under realistic preconditioning strategies?
- How do the results compare with state-of-the-art HPC implementations rather than desktop-scale assumptions?

The manuscript briefly discusses HPC considerations, but I believe this connection should be strengthened.

At minimum, I encourage the authors to justify their selected parameter ranges using representative scientific computing workloads.

More ambitiously, it would be highly valuable if the authors could identify one or more canonical application classes (for example, PDE-based simulation, finite-element methods, inverse problems, etc.) and evaluate their framework using parameter ranges motivated by those workloads.

Doing so would considerably improve the practical relevance of the reported quantum advantage boundaries.

Major Comment 4: Clarify audience and manuscript positioning

I found the intended audience somewhat unclear.

The introduction begins with a high-level discussion of quantum computing as a non-Von-Neumann paradigm beyond Moore's Law, which suggests an audience that may not be deeply familiar with quantum computing.

However, the technical portions of the manuscript rapidly transition into detailed discussions of HHL, Hamiltonian simulation, fault tolerance, and magic-state distillation.

The resulting narrative occasionally feels caught between an introductory article and a specialized technical resource-estimation paper.

I encourage the authors to consider more carefully:

- who the primary audience is,
- what prior knowledge is assumed,
- and how the introduction and framing can more naturally connect to the technical contributions.

Improving this narrative alignment would make the paper easier to follow and increase its impact.

Major Comment 5: Future work and conclusions

The discussion of future work raises many interesting questions. However, I felt that the conclusion section could be more focused and more clearly aligned with the manuscript's central contribution.

In particular, several future directions appear to follow naturally from the framework developed in the paper:

- application-driven benchmarking,
- hardware-independent energy modeling,
- sensitivity and uncertainty quantification,
- extension to additional quantum algorithms,
- identification of quantum advantage regions across diverse application domains.

I believe emphasizing these directions would strengthen the conclusion and further highlight the broader significance of the methodology introduced in the manuscript.